PanoEarth: Scaling Open-Vocabulary Panoptic Segmentation in Remote Sensing Images

Xinyang Pu, *Graduate Student Member, IEEE*, Jing Liu, *Member, IEEE*, Qiyu Chen, *Graduate Student Member, IEEE*, Hui Shen, and Feng Xu, *Senior Member, IEEE*

***Abstract*—Existing open-world remote sensing (RS) foundation models rely on text-to-image alignment for open-vocabulary semantic segmentation or object detection. However, semantic segmentation lacks instance-level delineation, while object detection fails to provide dense prediction for uncountable geographic features. Current approaches lack a unified solution for comprehensive panoptic understanding that jointly addresses semantic segmentation, instance segmentation, and object detection across all geographic elements. To bridge this gap, we propose PanoEarth, the first open-vocabulary panoptic segmentation framework for RS images, which integrates these tasks within a unified architecture. Our framework incorporates a scalable training strategy to compensate for missing background semantic information in instance segmentation data, enabling dense panoptic perception. Through the introduction of a domain-specific text embedding classifier to guide fine-tuning and establish precise mask query-to-category alignment, PanoEarth achieves competitive zero-shot open-vocabulary performance. We curate a large-scale panoptic dataset comprising 150k images spanning 111 geographic categories and millions of instances across landcover types, infrastructure elements, and fine-grained object subtypes. In zero-shot experiments, PanoEarth surpasses task-specific state-of-the-art models across multiple datasets and tasks. The joint co-training version enables unified extraction of 111 geographic categories, enabling hierarchical analysis from landcover mapping to fine-grained instance recognition. The source code is available at: https://github.com/xypu98/PanoEarth**

***Index Terms*—Open-vocabulary panoptic segmentation, remote sensing (RS), vision language model (VLM).**

I. Introduction

REMOTE sensing (RS) images contain complex and fine-grained foreground objects and diverse backgrounds, landcover types. The Earth observation through RS image interpretation requires open-world abilities on multiple tasks, such as semantic segmentation for landcover classification and instance segmentation for object identity. Panoptic segmentation [1] unifies these tasks by generating closed boundaries for objects and continuous masks for uncountable stuff, while open-vocabulary capability is enabled through vision-language models (VLMs) such as CLIP [2] with contrastive image–text pretraining. Open-vocabulary segmentation and detection [3] emerges to overcome the limitations of traditional closed-set models, enabling the flexible recognition and parsing of arbitrary categories beyond the predefined set.

Existing RS foundation models achieve open-world recognition by exploring the generalization abilities of VLMs. Several open-vocabulary semantic segmentation models [4], [5] demonstrating strong zero-shot performance in RS images, and open-world object detection models [6] based on DINO [7] excelling at fine-grained object localization. However, these approaches exhibit fundamental single-task limitations: semantic segmentation models produce pixel-level mask predictions but lack instance-aware delineation of object boundaries and countable quantities, while detection frameworks cannot characterize uncountable landcover features. This gap reveals a critical need for a unified open-vocabulary panoptic perception model that concurrently addresses both comprehensive landcover segmentation and instance-level object identification in Earth observation.

Panoptic perception in Earth observation faces significant challenges due to the lack of comprehensive annotations that simultaneously capture land cover categories and countable objects, which hinder the scalability of RS panoptic segmentation models. Current RS panoptic segmentation datasets [8], [9] remain limited in both scale and diversity. While landcover segmentation datasets [10], [11], [12] provide semantic masks for common terrain types, they lack annotations for discrete objects. Conversely, instance segmentation datasets [13] offer detailed object masks but treat most landcover pixels as unlabeled background. This annotation gap causes two key issues: inadequate training data for robust panoptic learning, and severe performance drops during mixed-dataset training, leading to either overfitting or incoherent predictions.

To bridge the gaps in models and data, we present PanoEarth, the first open-vocabulary panoptic segmentation model for the RS community. Our framework simultaneously addresses semantic segmentation, instance segmentation, and object detection through an end-to-end architecture with multitask segmentation datasets. We design a scalable training strategy to compensate for the absence of a staff label for instance segmentation data. In this way, PanoEarth is able to provide dense panoptic prediction and achieve robust open-vocabulary abilities.

Received 21 November 2025; revised 8 February 2026; accepted 5 March 2026. Date of publication 10 March 2026; date of current version 17 March 2026. This work was supported by the National Natural Science Foundation of China under Grant W2411057. *(Corresponding authors: Jing Liu; Feng Xu.)*

The authors are with the Key Laboratory for Information Science of Electromagnetic Waves (Ministry of Education), School of Information Science and Technology, Fudan University, Shanghai 200433, China, and also with Zhejiang Laboratory, Hangzhou 311100, China (e-mail: liujinglj@zhejianglab.edu.cn; fengxu@fudan.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3672643

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. (Left) PanoEarth integrates multitask datasets with a scalable training strategy. (Middle) Zero-shot result visualization of PanoEarth on the open-vocabulary panoptic segmentation task. (Right) PanoEarth surpasses task-specific state-of-the-art models on multiple evaluation metrics across five datasets and three tasks.

In addition, CLIP-based open-vocabulary models employ vision-language alignment for category-agnostic segmentation. However, RS imagery presents different stuff and instance characteristics compared to the natural images used for foundation model pretraining. This suggests that adapting pretrained models may benefit particularly from optimizing domain-specific text-embedding-to-query alignment during fine-tuning. PanoEarth accordingly incorporates an additional CLIP text encoder pretrained on RS images as a domain classifier to capture category characteristics. The dual-classifier fusion better aligns foundation models with RS text embeddings. The overhead perspective yields more consistent visual–textual distributions than natural images, enabling more precise vision-language matching for geographical objects.

To address the lack of comprehensive panoptic datasets in RS, we establish "stuff" categories for landcover features and "things" for countable objects. We then construct the PanoRS-C111 dataset by reorganizing 12 public datasets, creating a unified collection of 84 things and 27 stuff categories (111 total) to advance RS panoptic segmentation research.

We evaluate PanoEarth on the PanoRS-C111 dataset through comprehensive experiments, including zero-shot open-vocabulary tests on the subsets and all-data joint co-training. Our model demonstrates superior zero-shot performance, outperforming state-of-the-art segmentation and detection methods across five unseen datasets in landcover, instance segmentation, and object detection tasks. Through large-scale co-training on all 111 PanoRS-C111 categories, we develop PanoEarth+, a unified model achieving robust panoptic segmentation. Comparative results with task-specific SOTA models are visualized in Fig. 1.

Our contributions are threefold.

1) We propose PanoEarth, the first open-vocabulary panoptic segmentation model in RS, and achieve end-to-end joint co-training of multitask datasets and inference. A domain CLIP text embedding serves as a guided classifier to optimize the alignment between text embeddings and mask queries in RS images.
2) We design a scalable training strategy to achieve dense panoptic prediction by compensating extra stuff information for instance segmentation data to avoid foreground overfitting. We curate the PanoRS-C111 dataset that includes 84 countable things and 27 uncountable stuff categories.
3) Comprehensive experiments demonstrate that PanoEarth achieves competitive zero-shot performance across five RS datasets, evaluated by multitask metrics.

The remainder of this article is structured as follows. Section II provides a comprehensive review of related work on PanoEarth. Section III details the architecture of the proposed method. Section IV introduces the curated PanoRS-C111 dataset. Section V presents an extensive analysis of quantitative and qualitative results, including comparisons of multiple experiments—such as zero-shot evaluation and co-training—along with ablation studies. Finally, Section VI concludes this article.

II. Related Works

A. Panoptic Segmentation in RS

Panoptic segmentation unifies semantic and instance segmentation for comprehensive scene parsing. While early dual-branch architectures like Panoptic FPN [1] and UPSNet [14] set the foundation, transformer-based frameworks such as MaskFormer [15], Mask2Former [16], and OMG-Seg [17] now achieve universal segmentation via unified mask prediction. Furthermore, EOV-Seg [18] introduces language-guided open-vocabulary capabilities for arbitrary category segmentation. Recently, RS interpretation works [19], [20] have enhanced multimodal feature extraction. However, general models often struggle with the unique geometric and multiscale distributions of bird's-eye RS imagery. Our method bridges this gap by optimizing feature extraction specifically

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

for RS domain knowledge, addressing the limitations of natural-scene-centric approaches.

Panoptic segmentation has been increasingly adapted for RS imagery to address domain-specific challenges. Spatial-semantic consistency learning [21] and cross-view matching [22] capture RS spatial priors, while MFGS2Net [23] and MSIAF [24] focus on morphological guidance and feature interaction for multiscale object separation. Hybrid approaches, such as extending semantic models with nonlearning postprocessing [25], and the hierarchical fusion strategy of PiV-AHPC [26], demonstrate the efficacy of specialized architectures for high-dimensional RS data. Data-driven progress is supported by datasets like PASTIS [27], BSB [8], and FineGrip [9], though limitations in image scale and category diversity persist. Existing RS models remain hindered by data scarcity and closed-set constraints. To address this, we leverage RS-specific vision-language alignment and a scalable data-building strategy, enhancing open-vocabulary robustness and generalization.

B. Open-Vocabulary Segmentation and Detection

Open-vocabulary detection and segmentation leverage VLMs and large-scale pretraining to recognize categories beyond predefined label sets. In detection, ViLD [28] adapts CLIP for unseen categories, while RegionCLIP [29] enables fine-grained region-text alignment. GLIP [30] and Grounding DINO [31] further unify object detection with phrase grounding to enhance zero-shot transfer via multimodal fusion [7]. Parallelly, segmentation frameworks like OpenSeg [32] combine mask proposals with CLIP classification. GroupViT [33] enables segmentation emergence through text supervision, and MaskCLIP [34] utilizes masked self-distillation for dense prediction. Universal models, such as OpenSeeD [3] and OMG-Seg [17], demonstrate the feasibility of unifying these tasks. However, general models' reliance on exhaustive annotations induces foreground bias in label-sparse RS domains, undermining the dense scene parsing essential for Earth observation.

Focusing on RS, LAE-DINO [6] utilizes dynamic vocabulary construction and visual-guided prompt learning for robust detection. Dual-perspective alignment [35] further refines multimodal fusion for RS object detection, while GSNET [36] combines RS-specific priors with generalist CLIP features for flexible segmentation. SegEarth-OV [4] improves deep feature spatial recovery via a training-free approach, and SkySense-O [5] integrates vision-centric modeling for open-world interpretation. Unlike task-specific RS models that decouple instance detection from land-cover parsing, PanoEarth employs a scalable joint training strategy. By unifying foreground and background proposals into a shared representation, our method overcomes data constraints and achieves more comprehensive scene interpretation.

III. Methodology

PanoEarth introduces an open-vocabulary panoptic segmentation architecture for RS images. After introducing the formulation of the panoptic segmentation task, the proposed method is depicted in four key components: 1) an end-to-end panoptic segmentation network unifying semantic and instance segmentation; 2) open-set category matching using dual CLIP text embeddings enhanced with domain-specific knowledge; 3) joint loss optimization and open-vocabulary inference; and 4) a scalable training strategy for dense panoptic prediction.

A. Task Formulation of Panoptic Segmentation

RS panoptic segmentation aims to provide a unified spatial interpretation of Earth observation scenes. Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, the task assigns each pixel i a semantic label $l_i \in \mathcal{C}$ and a unique instance ID $z_i \in \mathbb{N}$. The label set $\mathcal{C}$ is partitioned into two disjoint subsets: $\mathcal{C}_{stuff}$ (amorphous background regions, e.g., water) and $\mathcal{C}_{thing}$ (countable foreground objects, e.g., ships). For pixels in $\mathcal{C}_{stuff}$, z_i is typically ignored, whereas for $\mathcal{C}_{thing}$, the same (l_i, z_i) pair identifies a specific physical entity.

PanoEarth reformulates this task as a set-prediction problem, representing the output as a collection of N candidate segments $\mathcal{Y} = \{(p_n, m_n)\}_{n=1}^{N}$. Each segment n is defined by

$$p_n \in \Delta^{K+1}, \quad m_n \in [0,1]^{H\times W} \tag{1}$$

where p_n denotes the probability distribution across K categories plus a "no object" label ($\emptyset$), and m_n represents the corresponding binary mask.

The final panoptic result for each pixel i is inferred by identifying the segment index n^* that maximizes the class confidence while exceeding a spatial threshold τ

$$n^* = \underset{n:m_n[i]>\tau}{\arg\max}\, p_n(c), \quad c \in \mathcal{C}. \tag{2}$$

This formulation ensures that each pixel is assigned to a unique, highly confident segment, enabling holistic scene parsing without complex postprocessing.

B. Architecture Overview

As illustrated in Fig. 2, PanoEarth adopts an simple architecture consisting of three key components: a frozen CLIP-pretrained backbone for visual feature extraction, a lightweight mask decoder following Mask2Former to generate category-agnostic mask proposals, and a dual open-vocabulary classifiers employing text embeddings from two CLIP text encoders that compute joint cosine similarity matching with object queries for category prediction.

1) Frozen Backbone From VLM Visual Encoder: The CLIP-based VLMs, pretrained on natural scenes, exhibit fundamental feature extraction capabilities and transferable generalization to RS images. In PanoEarth, the vision encoder remains frozen during training to preserve generalization capacity while accelerating the training process. The visual backbone employs the ConvNeXt [37] architecture, which modernizes conventional ConvNets through hierarchical Transformer-inspired designs. It features a four-stage hierarchical structure with progressively reduced spatial resolution and expanded channel dimensions $\{d_0, d_1, d_2, d_3\}$. Given an input image $\mathbf{I}$, the visual backbone $\mathbf{E}_{visual}$ extracts visual features $\mathbf{F} \in \mathbb{R}^{H'\times W'\times d}$. The final three hierarchical features $\{\mathbf{F}_i\}_{i=1}^{3}$ with dimensions $\{d_1, d_2, d_3\}$ are subsequently fused by the decoder for further processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

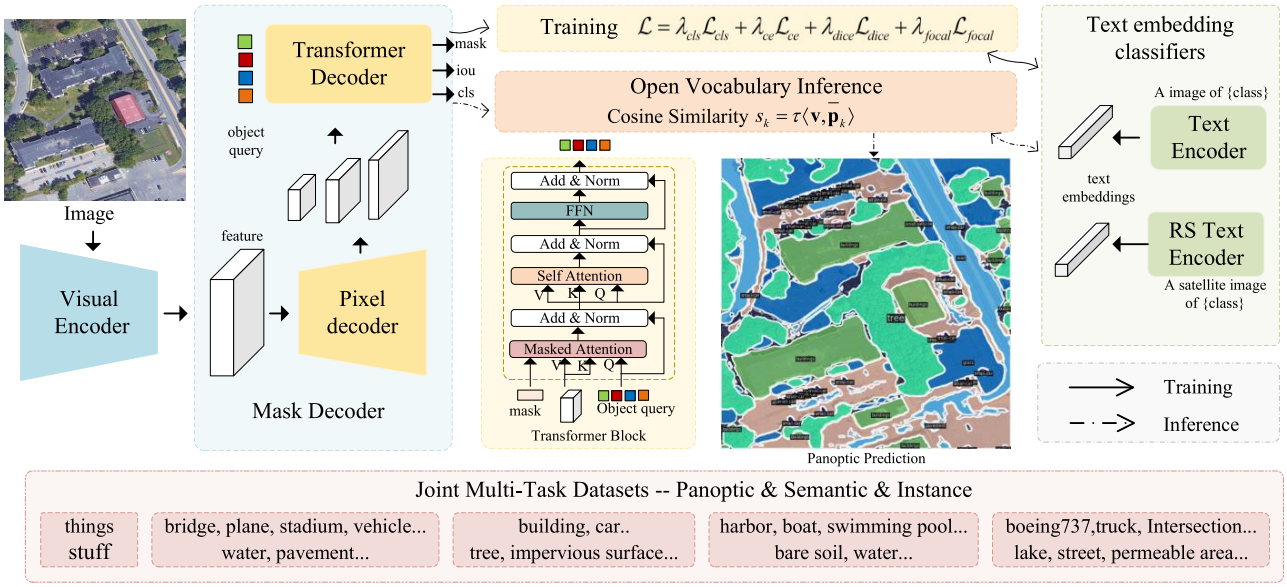


Fig. 2. Architecture of PanoEarth contains a visual encoder, a mask decoder, and two open-vocabulary classifiers. The joint co-training integrates multitask datasets with diverse categories. The open-vocabulary inference predicts categories by computing the cosine similarity of object query and text embeddings.

2) Tuning Mask Decoder for Category-Agnostic Query: The mask decoder architecture in PanoEarth, comprising a pixel decoder and a transformer decoder, extends Mask2Former with specialized modifications for category-agnostic mask prediction, mapping the set of object queries to the N candidate segments $\mathcal{Y}$.

The pixel decoder employs deformable attention to hierarchically integrate multiscale frozen features $\{\mathbf{F}_i\}_{i=1}^{3}$, where i indexes progressively finer spatial resolutions, and obtains the fused feature $\mathbf{F}_{fused}$. Within this architecture, a set of N learnable object queries $\mathbf{q}_j$ undergo refinement through masked cross-attention, where they serve as queries interacting with the fused features $\mathbf{F}_{fused}$ as keys and values to capture potential thing or stuff regions. These refined queries are further processed by multihead self-attention before computing the final object queries $\mathbf{q}_r$ through dot-product with the highest resolution fused features, effectively generating the binary masks m_n for each segment. The final refined object queries $\mathbf{q}_r \in \mathbb{R}^d$ are responsible for computing category matching scores p_n with the text embedding classifiers. This unified query-to-segment decoding ensures that each query index $j \in \{1, \ldots, N\}$ corresponds to a unique candidate in the panoptic output.

C. Dual-Text Embedding Classifiers Supervision

Standard CLIP text embeddings typically function as classifiers for open-vocabulary segmentation by computing cosine similarity between visual and textual features. However, the distinct overhead perspective of RS images creates semantic disparities with natural images, causing suboptimal performance when using natural scene domain-trained embeddings. To resolve this, we propose a dual-classifier approach incorporating both a general image-pretrained encoder E^g_{text} and a domain-specific RS encoder E^d_{text}, enabling domain-adaptive feature alignment during fine-tuning.

For each category $c_k \in \mathcal{C}$, our framework constructs two open-vocabulary classifiers through prompt-based embedding generation. In particular, multiple textual templates (e.g., $\mathcal{T}^g_k$ for general "a photograph of {cls}" and $\mathcal{T}^d_k$ for RS domain "a satellite image of {cls}") are encoded into normalized embeddings $\bar{\mathbf{e}}^i_k = \mathbf{e}^i_k / \|\mathbf{e}^i_k\|_2$ via each text encoder. Category prototypes are then computed as $\mathbf{p}_k = 1/|\mathcal{T}_k| \sum_i \bar{\mathbf{e}}^i_k$ and normalized to unit norm $\bar{\mathbf{p}}_k = \mathbf{p}_k / \|\mathbf{p}_k\|_2$.

The mask decoder's output queries $\mathbf{q}_r \in \mathbb{R}^d$ are processed through a three-layer FFN and projected onto the same metric space $\mathbf{v}$, enabling two classification scores P_g and P_d to be computed through cosine similarity $s_k = \tau\langle \mathbf{v}, \bar{\mathbf{p}}_k \rangle$ with a learned temperature τ, followed by softmax normalization $P(y = k|\mathbf{v}) = \exp(s_k)/\sum_j \exp(s_j)$ for both classifiers. The final category prediction P_{cls} combines both outputs through parameterized fusion

$$P_{cls} = \alpha P_g + (1 - \alpha) P_d \tag{3}$$

where α balances their contributions, thereby explicitly enforcing domain alignment between natural scenes and RS distributions to enhance open-vocabulary robustness.

D. Unified Training and Open-Vocabulary Inference

PanoEarth conducts an end-to-end framework where both training and inference utilize dual CLIP text embeddings for classification. During training, the model optimizes through Hungarian matching between predicted queries and ground-truth masks, minimizing the composite loss

$$\mathcal{L} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{ce}\mathcal{L}_{ce} + \lambda_{dice}\mathcal{L}_{dice} + \lambda_{focal}\mathcal{L}_{focal} \tag{4}$$

where $\mathcal{L}_{cls}$ represents the cross-entropy classification loss computed on fused predictions from both classifiers. For mask refinement, $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$ denote mask binary cross-entropy and dice losses, respectively. For mask refinement, $\mathcal{L}_{ce}$ and $\mathcal{L}_{dice}$ denote binary cross-entropy and dice losses, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

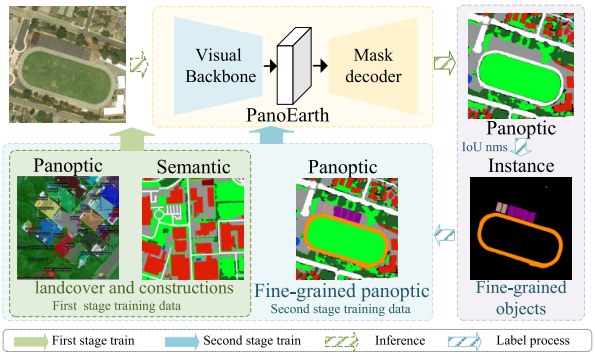


Fig. 3. Pipeline of scalable training strategy to provide dense panoptic annotations by compensating missing background annotations and conducting NMS with an IOU threshold for instance segmentation data.

TABLE I
HIERARCHICAL PRIORITY RULES FOR LABEL FUSION

Priority	Fusion Logic and Conflict Resolution
Level 1	Instance GT: Highest priority. Always retained. Overlapping regions from lower levels are discarded.
Level 2	Things Pseudo-labels (building): Assigned if no GT exists. If IoU $\geq$ 0.5 with GT, the area is merged into Level 1 via NMS.
Level 3	Stuff Pseudo-labels (e.g., road, water): Lowest priority. Only filled in pixels devoid of Level 1 and 2 annotations.

$\mathcal{L}_{ce}$ supervises pixel-level classification accuracy for fine-grained details, while $\mathcal{L}_{dice}$ optimizes regional overlap to ensure the geometric quality and IoU of the predicted masks. $\mathcal{L}_{focal}$ is the binary focal loss utilized for mask prediction to mitigate the extreme pixel-level imbalance between small-scale foreground objects and extensive background regions in RS imagery.

For inference, open-vocabulary categories are processed through the same prompt templates, computing final predictions via joint cosine similarity between object queries and two text embeddings, following the same fusion strategy as training.

E. Scalabe Training Strategy for Dense Prediction

Public RS datasets typically specialize in either dense land-cover semantic segmentation or sparse foreground instance segmentation. This disparity creates a data imbalance between large-area "stuff" (e.g., vegetation and water) and localized "things" (e.g., planes and cars) surrounded by vast unlabeled regions. Such fragmentation hinders multitask co-training, impairing the model's ability to simultaneously perform dense land-cover prediction and fine-grained instance extraction.

As shown in Fig. 3, we propose a scalable training pipeline to address the label sparsity and incompleteness in existing RS datasets. Leveraging PanoEarth's multidataset joint training design and its class-agnostic mask decoder, we design a two-stage training process detailed in Algorithm 1 and Table I. First, we train the model on densely annotated panoptic and semantic segmentation data to jointly learn large-area stuff classes and building-scale things instances, ensuring robust feature extraction for both. Next, we infer

Algorithm 1 Pseudo-Label Fusion for Scalable Training

1: Input:
- Densely labeled datasets $\mathcal{D}_{dense}$ (panoptic/semantic)
- Sparsely labeled instance datasets $\mathcal{D}_{sparse}$
- Pretrained model M_{init} with class-agnostic mask decoder

2: Stage 1: Dense Feature Learning

1) Train M_{init} on $\mathcal{D}_{dense}$ to jointly learn:
- Large-area *stuff* classes (vegetation, water, etc.)
- Building-scale *things* instances

2) Output: Robust feature extractor M_{dense}

3: Stage 2: Pseudo-Label Fusion

1) For each image $I \in \mathcal{D}_{sparse}$:
- Generate pseudo-labels $\mathcal{M}_{pred}$ using M_{dense}:
- *stuff* regions in unlabeled areas $\mathcal{M}_{pred_{stuff}}$
- *things* instances (building)$\mathcal{M}_{pred_{things}}$
Fuse with ground-truth instance labels via

$$\text{Let } U = \mathcal{M}_{pred_{things}} \cup \mathcal{M}_{gt}$$
$$\text{Then } \mathcal{L}_{fused}(p) =$$
$$\begin{cases} \mathcal{L}_{gt}(p) & \text{if } p \in U \text{and IoU} > 0.5 \\ \mathcal{M}_{pred}(p) & \text{otherwise} \end{cases}$$

- Apply class-agnostic NMS to resolve overlaps

2) Retrain model on $\mathcal{L}_{fused}$ data with reinitializition:
- Training from pre-trained weights M_{init}

4: Output: Unified panoptic segmentation model M_{final} for:
- Dense landcover prediction (*stuff*)
- Fine-grained instance extraction (*things*)

pseudo-labels for missing stuff regions and building objects on sparsely annotated instance segmentation datasets, then fuse them with the original instance labels. During fusion, instance labels are prioritized: any predicted things masks overlapping with ground-truth instance annotations (IoU >0.5) are replaced via IoU-based nonmaximum suppression (NMS), while stuff pseudo-labels supervise remaining unassigned regions.

The second training phase utilizes refined pseudo-labels with instance categories while reinitializing with original pre-trained weights, maintaining consistent feature extraction. Our category-agnostic approach—where masks are decoded independently of class predictions via cosine similarity—prevents category conflicts while preserving instance accuracy. This enables simultaneous thing and stuff segmentation without annotation overfitting, yielding dense panoptic masks.

IV. PANORS-C111 DATASET

We curate a comprehensive panoptic segmentation dataset named PanoRS-C111 of RS images by integrating and reformatting 12 public RS segmentation datasets to support panoptic segmentation tasks, including BSB Aerial Dataset [8], LOVEDA [10], DeepGlobe [11], OEM [38], ISPRS Potsdam [12], ISPRS Vaihingen, FloodNet [39], FLAIR [40], SOTA [13], and SIOR, FAST, and iSAID [41].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPOSITIONAL ANALYSIS OF SOURCE DATASETS WITH THINGS AND STUFF CATEGORY LISTS

Panoptic Segmentation Datasets					
Source Dataset	Total Classes	Things Classes	Things List	Stuff Classes	Stuff List
BSB	15	12	swimming pool, harbor, vehicle, boat, sports court, soccer field, commercial building, commercial building block, residential building, house, small construction	3	street, permeable area, lake

Semantic Segmentation Datasets					
Source Dataset	Total Classes	Things Classes	Things List	Stuff Classes	Stuff List
LoveDA	6	1	building	5	road, water, agriculture land, forest land, barren land
OEM	8	1	building	7	bare land, grass, pavement, road, tree, water, agriculture land
DeepGlobe	6	0	–	6	grass, tree, water, agriculture land, barren land, urban land
Potsdam	5	2	building, car	3	impervious surface, low vegetation, tree
Vaihingen	5	2	building, car	3	impervious surface, low vegetation, tree
FloodNet	9	4	building-flooded, building-non-flooded, vehicle, pool	5	road-flooded, road-non-flooded, water, tree, grass
FLAIR	12	1	building	11	pervious-surface, impervious-surface, bare soil, water, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, agricultural land, plowed land

Instance Segmentation Datasets					
Source Dataset	Total Classes	Things Classes	Things List	Stuff Classes	Stuff List
SOTA	29	19	building, large-vehicle, swimming pool, helicopter, bridge, plane, ship, soccer-ball-field, basketball-court, ground-track-field, small-vehicle, baseball-diamond, tennis-court, roundabout, storage-tank, harbor, container-crane, airport, helipad	9	bare land, grass, pavement, road, tree, water, agriculture land, forest land, urban land
SIOR	31	21	building, bridge, plane, ship, basketball-court, ground-track-field, baseball-diamond, tennis-court, storage-tank, harbor, airport, chimney, expressway service area, expressway toll station, dam, golf field, overpass, stadium, train station, vehicle, windmill		
FAST	52	38	A220, A321, A330, A350, ARJ21, Baseball-Field, Basketball-Court, Boeing737, Boeing747, Boeing777, Boeing787, bridge, bus, C919, cargo-Truck, dry-cargo-Ship, dump-truck, engineering-ship, excavator, fishing-boat, football-field, intersection, liquid-cargo-ship, motorboat, other-airplane, other-ship, other-vehicle, passenger-ship, roundabout, small-car, tennis-court, tractor, trailer, truck-tractor, tugboat, van, warship, building	13	bare land, grass, pavement, road, tree, water, agriculture land, forest land, urban land, permeable area, lake, pervious surface, bare soil
iSAID	30	16	ship, store tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, harbor, building		
Total	111	84	–	27	–

By classifying stuff for uncountable landcover categories, such as water, vegetation, and road, typically addressed in landcover semantic segmentation, and things for countable, fine-grained instances including buildings, vehicles, aircrafts, ships, and man-made structures, the PanoRS-C111 dataset encompasses a total of 111 categories (84 things and 27 stuff) and 150k images. Table II provides quantitative details about the constituent datasets, including their original tasks, category distributions, and stuff/things partitions. This integration enables large-scale panoptic segmentation while maintaining backward compatibility with original task formulations.

The implementation in PanoEarth facilitates multidataset co-training through independent yet unified annotation preprocessing for each constituent dataset. Original panoptic segmentation datasets require no additional processing, while instance segmentation datasets—containing only "things" categories with unique instance identifiers—undergo simpler processing compared to semantic segmentation data. The conversion protocol from semantic to panoptic segmentation comprises three key steps.

1) *Category Specification:* Explicitly designate "things" (c_t categories) and "stuff" (c_s categories) for each dataset, where c_t and c_s denote the counts of things and stuff categories, respectively.
2) *Label Remapping:* Reindex segmentation labels such that things category IDs precede stuff categories, resulting in contiguous class IDs ranging from 0 to $c_t + c_s - 1$ for all $c_t + c_s$ categories.
3) *Instance Encoding:* For things categories.
    1) Extract mask contours using closed boundary detection.
    2) Treat each connected region as a distinct instance.
    3) Generate panoptic IDs via bitwise operation: panoptic_id = (category_id $\ll$ 17) + instance_id.

    For stuff categories:panoptic_id = category_id $\ll$ 17
4) Visual Encoding: Apply ID-to-RGB pseudocolor transformation to preserve instance information in trainable label formats. Fig. 4 shows the visualization examples

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

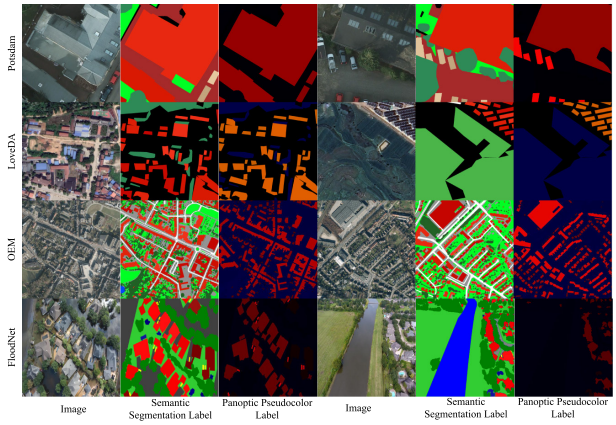


Fig. 4. Visualization of semantic segmentation labels and corresponding panoptic pseudocolor labels of PanoRS-C111.

of semantic segmentation mask labels and their corresponding panoptic pseudocolor mask labels of semantic segmentation datasets.

Instance datasets, already possessing instance IDs, bypass initial steps to directly generate panoptic IDs and pseudocolor labels. This standardized pipeline maps semantic labels to stuff and refines instance boundaries for things. While preserving original splits for benchmarking, we augment training data using our scalable strategy to produce dense panoptic annotations. To evaluate zero-shot performance, we curate specialized subsets for comparison with existing open-vocabulary RS image segmentation models. Finally, we release a robust open-vocabulary RS panoptic segmentation model trained on the complete PanoRS-C111 dataset.

V. Experiment Results and Analysis

A. Implementation Details

Our implementation initializes with pretrained weights [17], freezing the CLIP backbone to preserve generalization. Training employs 8× A100 80-GB GPUs with AdamW [42] ($lr = 1e^{-4}$, batch size = 8/GPU). Zero-shot experiments use 1-epoch fine-tuning, while PanoEarth+ co-training runs for 12 epochs. Dual-text embeddings fusion hyperparameter $\alpha = 0.8$. All experimental configurations utilize a resized input of 512×512 during both model training and evaluation.

B. Evaluation Metrics

The evaluation experiments employ the standard panoptic quality (PQ) metric, which is formulated as

$$PQ = \frac{\sum_{(p,g)\in TP} \mathrm{IoU}\,(p,g)}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|} \tag{5}$$

comprehensively analyzing panoptic segmentation accuracy for both stuff and things categories and classification precision. To thoroughly evaluate the multitask capabilities of PanoEarth and enable comparison with task-specific open-vocabulary models include object detection and semantic segmentation models, we additionally report mean intersection over union (mIoU) for semantic segmentation and box mAP (AP50) for instance segmentation and object detection.

C. Zero-Shot Open-Vocabulary Experiments

We conduct comprehensive experiments to evaluate PanoEarth's zero-shot open-vocabulary capabilities against state-of-the-art segmentation models [43], [44], [45], [46], and detection models. Trained on a curated subset of PanoRS-C111 (39 things and 12 stuff categories from seven datasets), we evaluate zero-shot performance across the other five unseen datasets encompassing both landcover and instance segmentation tasks.

For instance, in segmentation datasets, we employ open-vocabulary inference incorporating both things and stuff categories (listed in Table II) to prevent misclassification of background land-cover pixels as foreground instances, while strictly preserving the original ground-truth annotations (containing only instance labels) for fair evaluation. Other methods undergoes same inference procedures to ensure comparability. Although panoptic predictions include both semantic categories, evaluation metrics are computed exclusively on things classes in accordance with standard instance segmentation protocols. This method prevents artificial performance inflation from background labeling while maintaining rigorous evaluation standards, enabling comparison under identical conditions.

As shown in Table III, PanoEarth uniquely provides complete evaluation metrics for OVRSPS—a task where no comparable RS foundation models exist. Our model establishes state-of-the-art results: outperforming previous approaches by significant margins of 4.0 and 6.79 mIoU, respectively, on Potsdam and Vaihingen datasets for landcover segmentation while achieving comparable results to specialized models on iSAID instance segmentation. In addition, Table IV presents more comprehensive evaluation results of PanoEarth on multiple tasks, including panoptic segmentation (PQ, SQ, and RQ), instance segmentation(segmentation mAP, AP50, and AP75), object detection (bounding box mAP, AP50, and AP75) and semantic segmentation (mIOU, mF1, mPrecision, and mRecall).

Fig. 5 demonstrates PanoEarth's qualitative performance compared with state-of-the-art SkySense-O [5], where PanoEarth not only exhibits superior open-set capability but also achieves precise detection, segmentation, and boundary delineation for countable things instances, while simultaneously recovering missing background terrain information.

PanoEarth demonstrates superior dual capability: it matches the detection accuracy of LAE-DINO—the specialized OVRS SOTA—for fine-grained instances, while providing dense open-vocabulary parsing unavailable to dedicated detectors. Even in zero-shot settings, PanoEarth remains competitive with few-shot task-specific models.

Lower metrics in Tables III and IV relative to specialized baselines stem from two factors. First, while detectors like LAE-DINO optimize for fine-grained localization (e.g., FAST aircraft subtypes), PanoEarth prioritizes unified panoptic interpretation, balancing instance and category awareness. Second, the gap on FloodNet arises from perspective disparity; its oblique UAV imagery deviates from the nadir satellite perspectives used in PanoEarth's training. This tradeoff reflects a deliberate design choice, prioritizing cross-domain generality

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE III

ZERO-SHOT OPEN-VOCABULARY PERFORMANCE OF PANOEARTH ON MULTIPLE RS DATASETS AND COMPARISON WITH OPEN-SET MODELS. PQ, MAP(50), AND MIOU ARE EVALUATION METRICS OF PANOPTIC, INSTANCE, AND SEMANTIC SEGMENTATION TASKS, RESPECTIVELY

Model	Landcover Segmentation Datasets									Instance Segmentation Datasets					
	Potsdam			Vaihingen			FloodNet			iSAID			FAST		
	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU
Few-shot panoptic segmentation models															
OpenSeeD	20.84	–	29.89	10.94	–	23.17	10.91	–	20.89	–	–	–	4.41	–	5.34
OMG-Seg	36.62	55.10	54.42	19.56	36.40	25.18	**39.25**	9.02	40.11	4.76	5.00	17.84	2.09	0.01	5.22
Few-shot object detection models															
Grounding DINO	–	64.52	–	–	50.94	–	–	10.06	–	–	6.34	–	–	6.52	–
LAE-DINO	–	66.05	–	–	55.10	–	–	25.20	–	**–**	**36.21**	**–**	**–**	**22.13**	**–**
Few-shot semantic segmentation models															
SegGPT	–	–	30.12	–	–	47.27	–	–	33.81	–	–	29.17	–	–	**19.55**
DINOv	–	–	37.65	–	–	35.10	–	–	**44.55**	–	–	25.35	–	–	7.92
Zero-shot semantic segmentation models															
SAN	–	–	51.45	–	–	46.23	–	–	39.26	–	–	11.77	–	–	3.16
CAT-Seg	–	–	50.81	–	–	49.62	–	–	41.36	–	–	20.58	–	–	3.13
Zero-shot remote sensing semantic segmentation models															
GSNet	–	–	45.75	–	–	39.29	–	–	42.63	–	–	3.89	–	–	16.61
SegEarth-OV	–	–	47.10	–	–	29.10	–	–	34.00	–	–	21.70	–	–	2.84
SkySense-O	–	–	54.10	–	–	51.64	–	–	42.04	–	–	43.92	–	–	8.33
PanoEarth	**47.40**	**73.70**	**58.42**	**25.36**	**60.50**	**58.43**	26.48	**28.65**	40.61	**31.31**	26.55	**44.22**	**6.87**	5.32	7.52

TABLE IV

MULTITASK PERFORMANCE OF PANOEARTH ON ZERO-SHOT OPEN-VOCABULARY PANOPTIC SEGMENTATION EXPERIMENTS ACROSS DIFFERENT DATASETS. PQ: PANOPTIC QUALITY, SQ: SEGMENTATION QUALITY, RQ: RECOGNITION QUALITY, MAP: MEAN AVERAGE PRECISION, MIOU: MEAN INTERSECTION OVER UNION, AND MF1: MEAN $F1$-SCORE

Datasets	Panoptic Segmentation			Instance Segmentation			Object Detection			Semantic Segmentation			
	PQ	SQ	RQ	mAP	AP50	AP75	mAP	AP50	AP75	mIoU	mF1	mPrecision	mRecall
Potsdam	47.40	78.61	58.17	50.2	72.5	58.1	54.5	73.7	62.1	58.42	69.54	82.49	71.51
Vaihingen	25.36	72.29	34.52	31.6	59.8	28.2	36.2	60.5	37.8	58.43	72.55	70.72	76.90
FloodNet	26.48	60.77	32.64	14.0	26.9	13.5	17.5	28.6	17.4	40.61	51.04	55.09	56.00
iSAID	31.31	78.47	36.93	11.8	23.9	10.4	16.1	26.5	17.4	44.22	57.02	82.06	50.14
FAST	6.87	26.65	7.90	3.49	5.44	3.49	4.21	5.32	4.42	7.52	10.29	23.11	12.71

TABLE V

PERFORMANCE OF PANOEARTH+ JOINTLY TRAINED ON THE ENTIRE TRAINING SETS OF PANORS-C111 DATASET WITH 111 CATEGORIES AND EVALUATED ACROSS TEN LANDCOVER AND INSTANCE SEGMENTATION VALIDATION SETS (* INDICATES DATA BALANCED TRAINING)

Model	Landcover Segmentation Datasets														
	LoveDA			Potsdam			OEM			Vaihingen			FloodNet		
	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU
PanoEarth+	43.89	45.20	65.98	71.18	90.10	79.05	32.98	35.10	57.02	68.72	83.95	78.58	56.80	69.90	71.30
PanoEarth+*	52.59	53.40	71.74	72.34	91.30	81.28	38.40	38.20	62.61	71.06	86.20	81.07	59.70	74.15	75.02

Model	Instance Segmentation Datasets														
	BSB			FAST			SIOR			SOTA			iSAID		
	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU
PanoEarth+	69.94	61.05	71.67	40.20	37.10	47.75	50.98	46.82	57.53	49.79	43.60	48.62	61.87	55.90	82.3
PanoEarth+*	72.00	59.95	75.20	39.56	36.75	46.58	51.30	47.35	62.77	52.96	43.72	43.10	62.05	56.52	83.11

to ensure PanoEarth serves as a robust foundation model for broad-scale Earth observation.

D. Joint Cotraining Experiments

We also conduct large-scale joint co-training on the entire training sets of the PanoRS-C111 dataset, establishing PanoEarth+, a robust model capable of handling 111 categories. The proposed PanoEarth framework demonstrates exceptional scalability by unifying multiple vision tasks—including landcover classification, vegetation subtype recognition, building instance segmentation, flood detection, aircraft recognition, and warship identification—within a single joint training pipeline without requiring complex annotation mapping. As displayed in Table V, our joint

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.



Fig. 5. Qualitative comparison of zero-shot open-vocabulary panoptic segmentation results between PanoEarth and SkySense-O on representative RS scenes. Our method demonstrates superior instance boundary delineation (white boxes) and stuff region consistency, particularly for fine-grained object categories. For instance, segmentation datasets, we adopt open-vocabulary inference covering both things and 14 stuff categories in Table II to avoid background misclassification as foreground instances while maintaining original instance-only ground truth for evaluation.

co-training experiments are evaluated across ten validation sets, demonstrating state-of-the-art multitask performance.

E. Model Efficiency and Large-Scale Image Inference

The model comprises 221M parameters with 89% frozen and has a computational cost of 175 GFLOPs. For an input resolution of 512×512 pixels, the model achieves an average end-to-end latency of 903.12 ms on a single NVIDIA RTX 3090 GPU, which includes preprocessing, the model forward pass, and all postprocessing stages for dense instance processing. The memory consumption is 4786 MiB.

For high-resolution imagery (e.g., >2000 pixels), PanoEarth utilizes a sliding-window tiling strategy. Input images are partitioned into 512×512 patches with a stride of 500, processed independently, and reassembled into the global coordinate system. As shown in Fig. 6, this approach ensures seamless mask reconstruction and preserves object structural integrity across tile boundaries, regardless of image scale.

F. Ablation Studies

Under zero-shot open-set settings, we evaluate three critical components: continual fine-tuning preserves pretrained knowledge, scalable training addresses annotation gaps by incorporating pseudo-labels from initial landcover predictions, and a domain text embedding classifier provides semantic guidance for instance feature learning in RS images.

1) Necessity of Continual Fine-Tuning: Our experiments validate the critical role of continual fine-tuning in preserving pretrained knowledge while adapting to new domains. As shown in Table VI, the comparison between models loading backbone and mask decoder training from COCO panoptic task versus scratch-trained variants by only using GeoRSCLIP [48] backbone and initialing the mask decoder reveals two key findings. Pretrained models exhibit faster convergence, which only need 1 epoch learning due to retained generalization capacity; while the initialized decoder achieves 43.16 mIoU in the semantic segmentation task on the Potsdam dataset, its instance segmentation performance mAP achieves only 10.7,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF THREE KEY COMPONENTS ON POTSDAM, VAIHINGEN, AND FLOODNET DATASETS

Continual Fine-tuning	Scalable Training	Domain Classifier	Potsdam PQ	Potsdam mAP-box	Potsdam mIoU	Vaihingen PQ	Vaihingen mAP-box	Vaihingen mIoU	FloodNet PQ	FloodNet mAP-box	FloodNet mIoU
			2.50	9.80	37.17	1.62	6.00	32.27	1.58	2.25	28.53
	✓		17.15	10.75	43.16	13.20	6.30	47.95	18.70	1.75	31.36
✓			43.11	68.60	55.41	19.60	51.80	55.99	19.02	15.70	34.58
✓	✓		44.92	**74.10**	**59.31**	23.71	50.60	57.01	25.14	28.25	37.27
✓	✓	✓	**47.40**	73.70	58.42	**25.36**	**60.50**	**58.43**	**26.48**	**28.65**	**40.61**

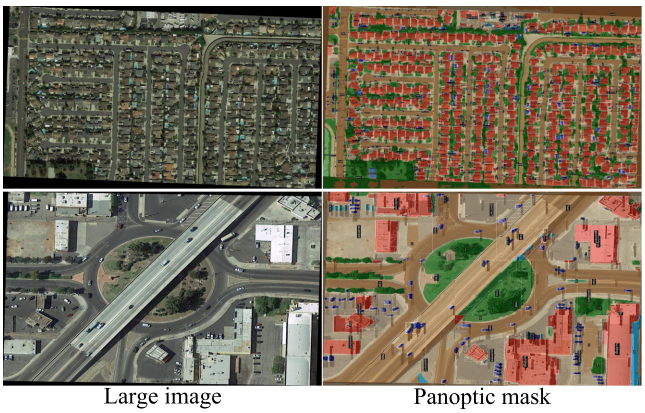


Fig. 6. Large-scale inference on DOTA [47] images (>2000 pixels). PanoEarth utilizes a 512 × 512 sliding-window strategy.

TABLE VII
QUANTITATIVE EVALUATION OF STAGE-1 PSEUDO-LABEL QUALITY ON OPTICAL IMAGES OF OPENEARTHMAP-SAR

Panoptic			Semantic			Instance
PQ	SQ	RQ	mIoU	Precision	Recall	$mAP50_{box}$
30.27	73.77	40.83	50.94	65.15	65.50	30.26

which is 63.4 lower than continual fine-tuning, exhibiting the necessity of continual fine-tuning.

2) Effectiveness of Scalable Training: Our scalable training strategy overcomes the challenge of incomplete background annotations in joint panoptic co-training. By intelligently propagating first-stage pseudo-labels to fill missing stuff regions, the method enhances prediction coherence. Quantitative results in Table VI show consistent gains, averaging 4.01 PQ and 2.54 mIoU across all datasets. Fig. 7 illustrates this completion process. While original instance labels contain only sparse things masks, our first-stage model generates comprehensive panoptic pseudo-labels. Notably, since the first stage is trained exclusively on building for the things category, it may coarsely misclassify other fine-grained objects (e.g., planes or vehicles) as building—a bias addressed during subsequent joint training.

To validate the reliability of stage-1 pseudo-labels, we conducted a quantitative evaluation on the OpenEarthMap-SAR dataset [49], which was excluded from training and contains optical RS images with eight categories similar to PanoRS. As shown in Table VII, the pseudo-labels achieve PQ of 30.27, mIoU of 50.94, and precision and recall reach 65.15 and 65.50, respectively, while the detection mAP is 30.26 for the things categories. These results confirm that the generated annotations provide a reliable supervisory signal for stage-2 joint training.

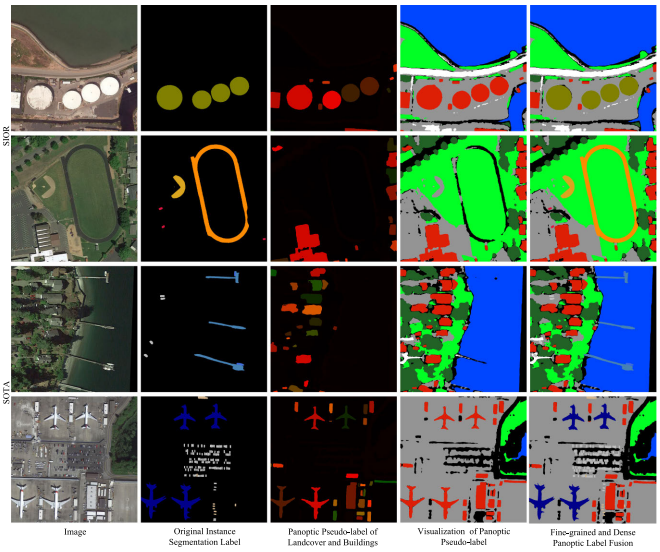


Fig. 7. Visualization of panoptic segmentation data. In pseudo-label masks, unique colors and IDs represent distinct instances, while dark regions indicate stuff with large ID values. In other visualizations, consistent colors denote the same category.

3) Analysis of Domain Text Embedding Classifier: Table VI validates the effectiveness of incorporating RS domain-specific CLIP text embeddings as auxiliary classifiers. The domain classifier demonstrates the improvements for fine-grained RS tasks, such as postflood area segmentation in FloodNet, achieving a 3.34 mIOU enhancement.

As shown in the T-SNE visualization of Fig. 8, while both ConvNeXt-L and ViT-L-14 successfully group semantically similar categories in Table VIII, they exhibit distinct clustering patterns in CLIP embeddings and complementary strengths in Table IX: ConvNeXt-L achieves a higher overall Silhouette score, indicating superior intercluster separation, whereas ViT-L-14 excels in intracluster compact-ness for specific classes, like aircraft and ground surfaces. This distributional divergence suggests that joint training could further enhance cross-domain feature representation for RS.

4) Balance of Multitask Datasets in Joint Cotraining: To resolve data imbalance between a small number of landcover datasets with dense labels and abundant instance data, we optimize joint training configurations. Table V demonstrates that a 5:1 semantic-to-instance oversampling ratio (PanoEarth+*)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

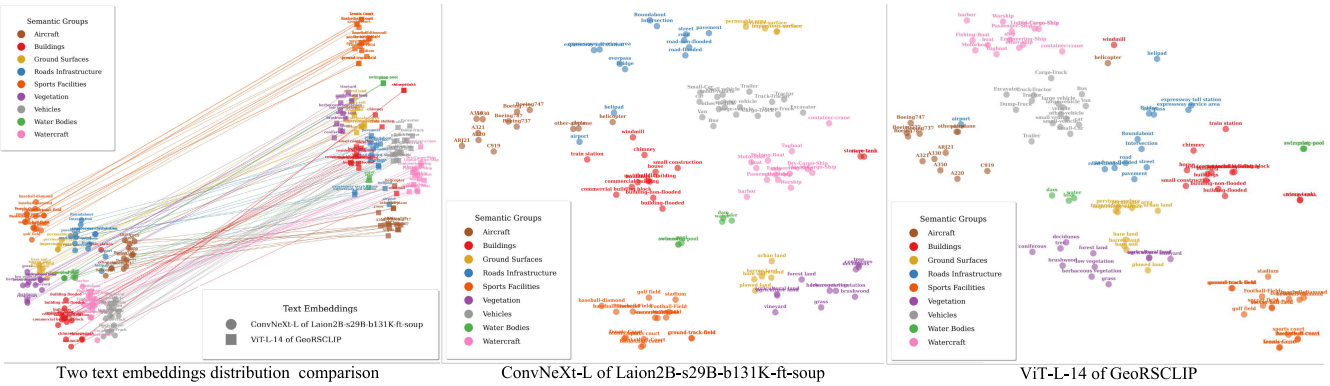


Fig. 8. T-SNE visualizations of the dual-text embeddings. (Left) Joint distribution alongside individual visualizations of general and (middle and right) RS-domain embeddings, respectively.

TABLE VIII

DETAILED CATEGORY GROUPING STRUCTURE IN FIG. 8 T-SNE VISUALIZATION. THE TABLE SHOWS NINE MAJOR SEMANTIC CLUSTERS AND THEIR CONSTITUENT FINE-GRAINED CATEGORIES

Cluster Name	Category Lists
Aircraft	helicopter, plane, A220, A321, A330, A350, ARJ21, Boeing737, Boeing747, Boeing777, Boeing787, C919, other-airplane
Buildings	buildings, building, commercial building, commercial building block, residential building, house, small construction, building-flooded, building-non-flooded, chimney, storage-tank, train station, store tank, windmill
Ground Surfaces	bare land, barren land, urban land, permeable area, pervious-surface, impervious-surface, impervious surface, bare soil, plowed land
Road Infrastructure	Bridge, roundabout, Roundabout, expressway service area, expressway toll station, overpass, road, road-flooded, road-non-flooded, street, pavement, Intersection, airport, helipad
Sports Facilities	soccer-ball-field, soccer ball field, Basketball-Court, basketball court, ground-track-field, ground track field, baseball-diamond, Baseball-Field, baseball diamond, tennis-court, Tennis-Court, tennis court, golf field, sports court, soccer field, Football-Field, ground-track-field, stadium
Vegetation	grass, tree, forest land, agriculture land, agricultural land, coniferous, deciduous, brushwood, vineyard, herbaceous vegetation, low vegetation
Vehicles	large vehicle, small vehicle, vehicle, car, Bus, Cargo-Truck, Dump-Truck, Excavator, Small-Car, Tractor, Trailer, Truck-Tractor, Van, other-vehicle, large-vehicle, small-vehicle
Water Bodies	swimming-pool, swimming pool, pool, water, lake, dam
Watercraft	ship, boat, Dry-Cargo-Ship, Engineering-Ship, Fishing-Boat, Liquid-Cargo-Ship, Motorboat, Passenger-Ship, Tugboat, Warship, other-ship, container-crane, harbor

TABLE IX

SILHOUETTE SCORES OF SEMANTIC CLUSTERS FOR TWO TEXT EMBEDDING MODELS. HIGHER VALUES INDICATE BETTER CLUSTER SEPARATION

Model	Overall	Aircraft	Buildings	Ground Surfaces	Roads Infra.	Sports Facilities	Vegetation	Vehicles	Water Bodies	Watercraft
ConvNeXt	0.276	0.212	0.162	0.229	0.078	0.412	0.141	0.468	0.408	0.341
ViT-L-14	0.241	0.274	0.140	0.321	0.053	0.359	−0.017	0.389	0.305	0.308

enhances stuff segmentation while preserving instance accuracy, enabling robust panoptic performance in RS images.

G. Stability and Reliability Analysis

To evaluate the robustness and precision of PanoEarth, we conduct a multidimensional analysis focusing on hyperparameter sensitivity, statistical stability, and boundary quality.

1) Hyperparameter Sensitivity Analysis: We evaluate the fusion weight α, which balances general semantic knowledge and RS priors, from 0.7 to 1.0. As shown in Table X, PanoEarth consistently achieves its peak PQ at $\alpha = 0.8$ across all three datasets (47.40 on Potsdam, 25.36 on Vaihingen, and 26.48 on FloodNet). While higher α values occasionally yield marginal gains in specific sub-metrics (e.g., mIoU on FloodNet at $\alpha = 0.9$), $\alpha = 0.8$ provides the most robust cross-dataset stability, effectively mitigating semantic misalignment in diverse Earth observation scenes.

2) Statistical Significance and Stability: To ensure the reliability of PanoEarth, we report performance as the *mean* $\pm$ *standard deviation* across two independent trials. As

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

TABLE X
EXPERIMENTAL RESULTS WITH DIFFERENT α VALUES

α	Potsdam			Vaihingen			FloodNet		
	PQ	mAP	mIoU	PQ	mAP	mIoU	PQ	mAP	mIoU
0.7	42.95	71.40	58.77	25.28	**62.20**	43.16	25.59	29.70	39.11
0.8	**47.40**	73.70	58.42	**25.36**	60.50	**58.43**	**26.48**	28.60	40.61
0.9	44.23	71.60	57.99	22.95	61.20	39.13	25.88	**33.20**	**42.51**
1.0	44.93	**74.10**	**59.31**	23.71	50.60	57.01	25.14	28.00	37.27

TABLE XI
QUANTITATIVE RESULTS ACROSS TWO INDEPENDENT RUNS

Potsdam			FloodNet		
PQ	mAP-box	mIoU	PQ	mAP-box	mIoU
47.39±0.01	73.60±0.10	58.42±0.00	26.45±0.03	27.33±1.33	39.16±1.45

TABLE XII
B-IOU RESULTS AT THREE PIXEL TOLERANCES ϵ

category	$\epsilon = 1$	$\epsilon = 3$	$\epsilon = 5$
things	27.31	50.93	62.65
stuff	14.80	27.34	35.11
all	23.75	44.21	54.81

summarized in Table XI, PanoEarth exhibits remarkable stability. For instance, the variance in PQ is as low as ±0.01 on Potsdam and ±0.03 on FloodNet. This consistently low variance across multiple runs demonstrates the high reproducibility and robustness of our scalable training strategy in open-vocabulary tasks.

3) Quantitative Boundary Evaluation: To quantitatively assess the mask precision, we evaluate the boundary IoU (B-IoU) at different pixel tolerances ($\epsilon \in \{1, 3, 5\}$). As shown in Table XII, PanoEarth demonstrates superior boundary adherence, particularly for things categories, achieving 62.65 at $\epsilon = 5$ and 50.93 at $\epsilon = 3$. Even under the most stringent tolerance ($\epsilon = 1$), the model maintains a B-IoU of 27.31 for instances. This high boundary fidelity confirms that our method effectively minimizes the and produces sharp, structurally sound masks for complex RS objects.

H. Failure Case Analysis on Fine-Grained Recognition

To further explore the limitations of PanoEarth in open-vocabulary scenarios, we present representative failure cases from the FAST dataset in Fig. 9. The quantitative results (6.87 PQ) and visual inspections reveal that the primary challenge lies in zero-shot fine-grained classification. While PanoEarth accurately localizes targets, it suffers from semantic aliasing, where fine-grained sub-categories (e.g., A350 are misclassified as generic airplanes due to highly clustered text embeddings in the latent space. In addition, instance merging occurs in densely packed scenes, such as harbors, where adjacent small objects are grouped into a single mask. These failure modes suggest that incorporating hierarchical taxonomic priors could further enhance the discriminative power for complex subtype categories in future work.

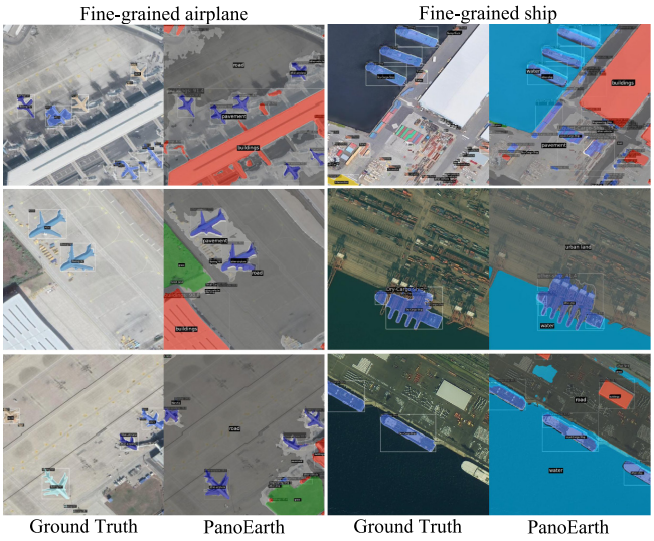


Fig. 9. Representative failure cases of PanoEarth on the FAST dataset. Ground truth only contains instance annotation.

VI. CONCLUSION

This article presents PanoEarth, the first open-vocabulary panoptic segmentation model in RS images. Our approach integrates landcover semantic segmentation and fine-grained instance segmentation through joint co-training within a scalable training strategy. The proposed domain embedding classifier enhances semantic learning for RS imagery, and the scalable training pipeline contributes to robust zero-shot generalization and dense prediction capabilities. We curate PanoRS-C111, a large-scale panoptic segmentation dataset with extensive foreground and background category definitions for a total of 111 classes. Comprehensive experiment results demonstrate that PanoEarth achieves competitive performance compared to existing task-specific state-of-the-art models. We believe this work offers a feasible paradigm and presents promising potential for unifying multiple visual tasks in open-world RS image interpretation.

REFERENCES

[1] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6392–6401.

[2] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, vol. 139, 2021, pp. 8748–8763.

[3] H. Zhang et al., "A simple framework for open-vocabulary segmentation and detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 1020–1031.

[4] K. Li et al., "SegEarth-OV: Towards training-free open-vocabulary segmentation for remote sensing images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 10545–10556.

[5] Q. Zhu et al., "SkySense-O: Towards open-world remote sensing interpretation with vision-centric visual-language modeling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 14733–14744.

[6] J. Pan et al., "Locate anything on earth: Advancing open-vocabulary object detection for remote sensing community," in *Proc. AAAI*, 2025, pp. 6281–6289.

[7] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9630–9640.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.

[8] O. L. F. de Carvalho et al., "Panoptic segmentation meets remote sensing," *Remote Sens.*, vol. 14, no. 4, p. 965, Feb. 2022.
[9] D. Zhao et al., "Panoptic perception: A novel task and fine-grained dataset for universal remote sensing image interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5620714.
[10] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "Loveda: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. NeurIPS Datasets Benchmarks*, vol. 21, 2021, pp. 1–12.
[11] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–181.
[12] J. Sherrah, "Fully convolutional networks for dense semantic labelling of high-resolution aerial imagery," 2016, *arXiv:1606.02585*.
[13] B. Du et al., "SAMRS: Scaling-up remote sensing segmentation dataset with segment anything model," in *Proc. Adv. Neural Inf. Process. Syst. 36*, 2023, pp. 8815–8827.
[14] Y. Xiong et al., "UPSNet: A unified panoptic segmentation network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8818–8826.
[15] B. Cheng, A. G. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. Neural Inf. Process. Syst.*, 2021, pp. 17864–17875.
[16] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1280–1289.
[17] X. Li et al., "OMG-seg: Is one model good enough for all segmentation?," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27948–27959.
[18] H. Niu, J. Hu, J. Lin, G. Jiang, and S. Zhang, "EOV-seg: Efficient open-vocabulary panoptic segmentation," 2024, *arXiv:2412.08628*.
[19] Z. Dong, Y. Gu, and T. Liu, "Generative ConvNet foundation model with sparse modeling and low-frequency reconstruction for remote sensing image interpretation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603816.
[20] D. Du, T. Liu, and Y. Gu, "A multimodal unified representation learning framework with masked image modeling for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5651216.
[21] Z. Dong, T. Liu, and Y. Gu, "Spatial and semantic consistency contrastive learning for self-supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote. Sens.*, vol. 61, 2023, Art. no. 5621112.
[22] R. Liu, T. Luo, S. Huang, Y. Wu, Z. Jiang, and H. Zhang, "CrossMatch: Cross-view matching for semi-supervised remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5650515.
[23] B. Sun et al., "MFGS2Net: A morphological features guided scale separation network for remote sensing panoptic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621717.
[24] Z. Sun, J. Liu, W. Zhang, F. Liu, J. Yang, and L. Xiao, "Multi-scale feature interaction and adaptive experts for panoptic segmentation in remote sensing images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2025, pp. 1–5.
[25] O. L. F. de Carvalho, O. A. de Carvalho Junior, A. O. de Albuquerque, N. C. Santana, and D. L. Borges, "Rethinking panoptic segmentation in remote sensing: A hybrid approach using semantic segmentation and non-learning methods," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[26] Y. Jiang, Y. Gu, and X. Li, "Pillar-voxel fusion network for 3D object detection in airborne hyperspectral point clouds," *Sci. China Inf. Sci.*, vol. 69, no. 1, Jan. 2026, Art. no. 112301.
[27] V. S. Fare Garnot and L. Landrieu, "Panoptic segmentation of satellite image time series with convolutional temporal attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4852–4861.
[28] X. Gu, T. Lin, W. Kuo, and Y. Cui, "Open-vocabulary object detection via vision and language knowledge distillation," in *Proc. ICLR*, 2022, pp. 1–20.
[29] Y. Zhong et al., "RegionCLIP: Region-based language-image pretraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16772–16782.
[30] L. H. Li et al., "Grounded language-image pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10955–10965.
[31] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. ECCV*, vol. 15105, 2024, pp. 38–55.
[32] M. Xu et al., "A simple baseline for open-vocabulary semantic segmentation with pre-trained vision-language model," in *Proc. ECCV*, vol. 13689, 2022, pp. 736–753.
[33] J. Xu et al., "GroupViT: Semantic segmentation emerges from text supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18113–18123.
[34] X. Dong et al., "MaskCLIP: Masked self-distillation advances contrastive language-image pretraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 10995–11005.
[35] Y. Liu, W. Guo, C. Yao, and L. Zhang, "Dual-perspective alignment learning for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5404015.
[36] C. Ye, Y. Zhuge, and P. Zhang, "Towards open-vocabulary remote sensing image semantic segmentation," in *Proc. AAAI*, 2025, pp. 9436–9444.
[37] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11966–11976.
[38] J. Xia, N. Yokoya, B. Adriano, and C. Broni-Bediako, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6243–6253.
[39] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89644–89654, 2021.
[40] A. Garioud, S. Giordano, N. David, and N. Gonthier, "FLAIR-HUB: Large-scale multimodal dataset for land cover and crop mapping," 2025, *arXiv:2506.07080*.
[41] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE CVPR Workshops*, Jun. 2019, pp. 28–37.
[42] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, Jun. 2019, pp. 1–19.
[43] X. Wang, X. Zhang, Y. Cao, W. Wang, C. Shen, and T. Huang, "SegGPT: Towards segmenting everything in context," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 1130–1140.
[44] F. Li et al., "Visual in-context prompting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 12861–12871.
[45] M. Xu, Z. Zhang, F. Wei, H. Hu, and X. Bai, "Side adapter network for open-vocabulary semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 2945–2954.
[46] S. Cho, H. Shin, S. Hong, A. Arnab, P. H. Seo, and S. Kim, "CAT-seg: Cost aggregation for open-vocabulary semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 4113–4123.
[47] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.
[48] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large-scale vision- language dataset and a large vision-language model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.
[49] J. Xia, H. Chen, C. Broni-Bediako, Y. Wei, J. Song, and N. Yokoya, "OpenEarthMap-SAR: A benchmark synthetic aperture radar dataset for global high-resolution land cover mapping," 2025, *arXiv:2501.10891*.



Xinyang Pu (Graduate Student Member, IEEE) received the B.E. degree in telecommunication engineering from Sun Yat-sen University, Guangzhou, China, in 2020. She is currently pursuing the Ph.D. degree in electronic science and technology with the Key Laboratory of Information Science of Electromagnetic Waves, Fudan University, Shanghai, China.

Her research interests include multimodal vision-language models and deep learning in remote sensing applications and intelligent interpretation of SAR images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.



Jing Liu (Member, IEEE) received the Ph.D. degree in information and communication engineering from Hainan University, Haikou, China, in 2020.

From September 2020 to 2022, she was a Post-Doctoral Researcher with Zhejiang Laboratory, Hangzhou, China, where she is currently a Senior Researcher. Her research interests include multimodal information processing, artificial intelligence, and machine vision. She has authored over 60 publications in these areas.

Dr. Liu is a member of the IEEE Geoscience and Remote Sensing Society (GRSS) Chapters Committee. She serves as a reviewer for IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE (TPAMI), IEEE TRANSACTIONS ON MULTIMEDIA (TMM), IEEE TRANSACTIONS ON MEDICAL IMAGING (TMI), *Information Fusion*, and major conferences, including AAAI, CVPR, IJCAI, and MICCAI.



Qiyu Chen (Graduate Student Member, IEEE) received the B.E. degree in intelligent science and technology from Fudan University, Shanghai, China, in 2025, where she is currently pursuing the Ph.D. degree in information and communication engineering with the Key Laboratory of Information Science of Electromagnetic Waves.

Her research interests include deep learning for remote sensing, embodied intelligence for perception, and decision-making tasks.



Hui Shen received the B.E. degree from the Shenyuan Honors College, Beihang University, Beijing, China, in 2017, and the M.E. degree from the School of Instrumentation and Optoelectronic Engineering, Beihang University, in 2020.

She is currently a Researcher with the Scientific Data Hub Research Center, Zhejiang Laboratory, Hangzhou, China. Her research interests include multimodal large models in remote sensing applications and vision foundation models.



Feng Xu (Senior Member, IEEE) received the B.E. degree (Hons.) in information engineering from Southeast University, Nanjing, China, in 2003, and the Ph.D. degree (Hons.) in electronic engineering from Fudan University, Shanghai, China, in 2008.

He is currently a Professor and the Vice Dean of the School of Information Science and Technology, Fudan University, and the Vice Director of the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Shanghai. His research interests include electromagnetic scattering modeling, SAR information retrieval, and radar system development.

Dr. Xu was a recipient of the Second-Class National Nature Science Award of China in 2011, the 2014 Early Career Award of the IEEE Geoscience and Remote Sensing Society, and the 2007 SUMMA Graduate Fellowship in the advanced electromagnetics area. He is the Founding Chair of the IEEE GRSS Shanghai Chapter and an IEEE GRSS AdCom Member. He is a Topic Associate Editor of IEEE TRANSACTIONS OF GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS from 2014 to 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 17:25:26 UTC from IEEE Xplore. Restrictions apply.